



GCE AS MARKING SCHEME

SUMMER 2019

**AS (NEW)
BIOLOGY - UNIT 2
2400U20-1**

INTRODUCTION

This marking scheme was used by WJEC for the 2019 examination. It was finalised after detailed discussion at examiners' conferences by all the examiners involved in the assessment. The conference was held shortly after the paper was taken so that reference could be made to the full range of candidates' responses, with photocopied scripts forming the basis of discussion. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners.

It is hoped that this information will be of assistance to centres but it is recognised at the same time that, without the benefit of participation in the examiners' conference, teachers may have different views on certain matters of detail or interpretation.

WJEC regrets that it cannot enter into any discussion or correspondence about this marking scheme.

WJEC GCE AS BIOLOGY(NEW)

UNIT 2 - BIODIVERSITY AND PHYSIOLOGY OF BODY SYSTEMS

SUMMER 2019 MARK SCHEME

GENERAL INSTRUCTIONS

Recording of marks

Examiners must mark in red ink.

One tick must equate to one mark (apart from the questions where a level of response mark scheme is applied).

Question totals should be written in the box at the end of the question.

Question totals should be entered onto the grid on the front cover and these should be added to give the script total for each candidate.

Marking rules

All work should be seen to have been marked.

Marking schemes will indicate when explicit working is deemed to be a necessary part of a correct answer.

Crossed out responses not replaced should be marked.

Credit will be given for correct and relevant alternative responses which are not recorded in the mark scheme.

Extended response question

A level of response mark scheme is used. Before applying the mark scheme please read through the whole answer from start to finish. Firstly, decide which level descriptor matches best with the candidate's response: remember that you should be considering the overall quality of the response. Then decide which mark to award within the level. Award the higher mark in the level if there is a good match with both the content statements and the communication statement. Award the middle mark in the level if most of the content statements are given and the communication statement is partially met. Award the lower mark if only the content statements are matched.

Marking abbreviations

The following may be used in marking schemes or in the marking of scripts to indicate reasons for the marks awarded.

cao	=	correct answer only
ecf	=	error carried forward
bod	=	benefit of doubt

Question			Marking details	Marks Available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	Eubacteria / Archaea	1			1		
		(ii)	Can {exist in / live in / thrive in / withstand/ survive} {extreme/ wide/ high/ owtte/ harsh} {environment/ conditions / temperature / pH / acidity / alkalinity / chemical concentration / pressure}	1			1		
		(iii)	{Plantae / plants} + Fungi + <u>Protocist(a)</u> (correct spelling only)	1			1		
	(b)	(i)	Phylum	1			1		
		(ii)	A. different (developmental/evolutionary) origin / don't share <u>recent</u> common ancestor (1) B. Analogous structures (1) C. Similar / same function (1)		3		3		
			Question 1 total	4	3	0	7	0	0

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			$3.6 \times 10^3 = 2$ marks If incorrect award 1 mark for $1.2 \times 10^3 \times 3 /$ $3600 /$ 36×10^2		2		2	2	
	(b)			Blood carried in {vessels / named vessels}/ vascularisation (1) transport {molecules / substances / materials / named substances} {to / from} {exchange surfaces/cells/ (named) tissues} (1) Ignore transported around body	2			2		
	(c)			Any four ($\times 1$) from A. Amphibian has <u>greater</u> SA:vol (ratio) / ora (1) B. Amphibian can use (permeable) skin for {gas exchange / diffusion/ to absorb oxygen}/ ora (1) C. Amphibians have short diffusion pathway/ ora (1) D. Amphibian has lower oxygen requirement / diffusion across skin alone would meet oxygen requirement/ ora (1) E. Amphibian has lower metabolic rate / lower body temperature/ does not need to maintain body temperature/ less energy is needed/ ora (1)		4		4		
				Question 2 total	2	6	0	8	2	0

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Less grazing of the clover/ avoid being eaten/ owtte Reject reference to predator		1		1		
	(b)	(i)		Any two (× 1) from Single leaf/ number of leaves (1) Size/ volume of tube (1) (Incubation) time (1) temperature (1)		2		2		2
		(ii)		Cyanide {released/ produced} / ora			1	1		1
		(iii)		(gets on to skin when) {handling paper/ touching paper} + use gloves/ forceps OR (Chemical) could be transferred from hand to eye when handling paper + wear eye protection/ wash hands Accept answer on either line e.g. use gloves when handling paper		1		1		1
	(c)	(i)		Independent = Altitude and Dependent variable = {number/ percentage} of cyanogenic clover	1			1		1
		(ii)		A. Axes/ tapes/ grid (1) B. random number table/calculator/dice/ other suitable method (to give coordinates) (1) C. {sample / collect} at intersect (of coordinates)/ from within quadrat/ owtte (1) reject count leaves	3			3		3
	(d)			No and Temperature not measured.			1	1		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)			Any two (× 1) from: A. The existence of {a number of / two / more than one/ several} {form(s) / traits/ types/ phenotypes} / or description of (1) B. that cannot be explained by mutation alone (1) C. {a number of / two / more than one/ several} alleles for the same {gene/ locus} (1)	2			2		
				Question 3 total	6	4	2	12	0	8

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			- Biconcave + {increases diffusion/ more oxygen absorbed/ increase gas exchange} / large surface area + {diffusion/ oxygen absorbed/ gas exchange} (1) {Biconcave/ thin} + short diffusion pathway (1) - Enucleate/no nucleus + more space for haemoglobin (1)	3			3		
	(b)			12 or 13 or 14 (μm) = 3 marks If incorrect award 2 marks for answer between 12-15 and shown to a decimal place If both above incorrect award 1 mark for (diameter of B/ diameter of A) $\times$ 8 (1)		3		3	3	
	(c)	(i)		100 (2) (45 / 4.5 $\times$ 10 ⁶) $\times$ 10 ⁷ (1)	1	1		2	2	
		(ii)		Yes. Mean volume higher than {normal / 95} (ecf)			1	1		
		(iii)		Any two ($\times$ 1) from: A. Fewer cells so {less haemoglobin/ lower total surface area} (1) B. Larger cell therefore {smaller SA : vol / increased diffusion pathway} (1) C. Larger cells so harder for red blood cells to pass through capillaries (1)			2	2		
				Question 4 total	4	4	3	11	5	0

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		2		1		1	1	
		(ii)		(-)1.7		1		1	1	
		(iii)		Lymph (capillary / vessel) / lymphatic Reject node/ lacteal	1			1		
	(b)			Any three (× 1) from A. Fewer plasma proteins/ not enough protein in blood (1) B. Which {raises/increases} {water/solute} potential of {plasma/blood}/ osmotic pressure is too low (1) C. reduces {water/solute} potential gradient / hydrostatic pressure is greater than osmotic pressure (1) D. less water/ fluid reabsorbed (by osmosis) (1) Ignore concentration of water			3	3		
				Question 5 total	1	2	3	6	2	0

Question			Marking details	Marks Available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)	{Large/long/sharp/pointed/curved} canines + {piercing/seizing/tearing/killing/ holding} (prey)		1		1		
		(ii)	Diastema + {manipulating food/owtte} OR {interlocking/WM} {molars/premolar} + {grinding/chewing} OR incisors + {slicing/cutting}		1		1		
	(b)		Duodenum mouth/buccal cavity stomach All correct = 2 marks 2 correct = 1 mark 1/0 correct = 0 marks	2			2		
	(c)		Any five (× 1) from: A. {shortening/blunting} of villi/ fewer <u>microvilli</u> (1) B. reduced SA (1) C. less {enzymes / named enzyme(s)} (1) D. so reduced {digestion / hydrolysis} (leading to weight loss)/ less absorption of {products of digestion/ named products}(1) E. more solutes in lumen / lower water potential in lumen (1) F. less water absorption / water moves from cells into lumen (leading to diarrhoea) (1)		1	4	5		
	(d)		A. Lowers Na ⁺ conc in epithelial cell / creates {conc/ diffusion} gradient for Na ⁺ (1) B. Na ⁺ and glucose enter (epithelial cell) by {cotransport / facilitated diffusion} (not active transport) (1) C. increases glucose conc in cell / creates {conc/diffusion} gradient for glucose (1) D. glucose {leaves epithelial cells/ enters blood} by <u>facilitated diffusion</u> (not active transport) (1)		4		4		
			Question 6 total	2	7	4	13		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			Chloroplasts (1) unevenly thickened cell walls (1)	2			2		
	(b)	(i)		Expressed to 3sf = 198 = 4 marks If incorrect sig fig allow 3 marks for 198.159 / 198.0758/ 198.16 If rounded area allow 3 marks for 197 / 200 If answer incorrect award 2 mark for calculation of area 0.07065 / 0.07068/ 0.071 If area incorrect award 1 mark for Substitution into formula ($\pi \times 0.15 \times 0.15$) = 1 mark *if candidate uses diameter instead of radius award 3 marks for 49.5 Award 2 marks for incorrect sig fig 49.539/ 49.522 Award 1 mark for calculation of area 0.2826 / 0.2827		4		4	4	4
		(ii)		Epidermal cells are larger {in shade/ diagram 2}/ORA (1) so fewer of them (per unit area) {in shade/ diagram 2}/ ORA (1)			2	2		
	(c)	(i)		Prevent {air/ bubbles} entering xylem (1) which would break {cohesion/ hydrogen bonds}/ affect cohesive forces (between water molecules) (1)	2			2		2

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		A. Humidity falls (1) B. because {ABA/ hormone} causes closure of stomata (1) C. causes reduced {transpiration/ water vapour loss/ evaporation} (from leaf/ stomata)(1) NOT stops D. (Benefit to plant) conserves water / reduces water loss / maintain turgor/ prevent wilting (when low soil water) (1) NOT stops Penalise stop once only		2	2	4		
				Question 7 total	4	6	4	14	4	4

Question	Marking details	Marks Available					
		AO1	AO2	AO3	Total	Maths	Prac
8	Indicative content Section A: From epidermis and across cortex: • Apoplast + symplast + vacuolar pathway • (Apoplast) Along cell walls • (Symplast) through cytoplasm/ plasmodesmata • (Vacuolar pathway) passes through tonoplast/ cytoplasm • Down water potential gradient / by osmosis Section B: Into vascular tissue: • Endodermis has Casparian strip • Waterproof / composed of suberin • Blocks apoplast pathway • Water {forced / passes} into symplast • Active transport of {salts / minerals} <u>into xylem</u> • Lowers water potential in <u>xylem</u> • Water enters <u>xylem</u> by osmosis / down water potential gradient Waterlogged soil: • {Less / no} oxygen (for respiration) • {Less / no} respiration so {less / no} {energy / ATP} • for active transport of minerals in the {endodermis/ root hairs} Must be in context of no oxygen • Lower water potential gradient as a result of less active transport of salts / minerals into xylem • Less water enters xylem which lowers root pressure	5	4				

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				7-9 marks Indicative content of this level is detailed content from all three areas <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is detailed content from two areas or less detail from three areas <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is any correct statement from indicative content <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question 8 total	5	4	0	9	0	0

UNIT 2: BIODIVERSITY AND PHYSIOLOGY OF BODY SYSTEMS**SUMMARY OF MARKS ALLOCATED TO ASSESSMENT OBJECTIVES**

Question	AO1	AO2	AO3	TOTAL MARK	MATHS	PRAC
1	4	3	0	7	0	0
2	2	6	0	8	2	0
3	6	4	2	12	0	8
4	4	4	3	11	5	0
5	1	2	3	6	2	0
6	2	7	4	13	0	0
7	4	6	4	14	4	4
8	5	4	0	9	0	0
TOTAL	28	36	16	80	13	12